Nikita Khvatkin Petrovsky

Predoctoral researcher. Radiation transport and machine learning for fusion facilities.

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Education

2025 – now **PhD, Doctoral Programme in Industrial Technologies**, UNED, Madrid
Provisional thesis title: Neural network Assisted CAd-based Real-Time dose-Estimator (NACARTE). Supervisors: Rafael Juárez Mañas and Pablo Martínez Albertos. Doctoral research work (TID) NACART^{2}E finished in September 2026.

2026 – now **Master in Computational Neutronics**, UNED, part time

2026 – now **Bachelor's Degree in Physics**, UNED, part time

2022 – 2024 **Master in Nuclear Science and Technology**, Universidad Politécnica de Madrid
Thesis: Modelling of TVA Watts Bar Unit 1 nuclear power plant with the Monte Carlo code OpenMC (February 2024). Supervisor: Emilio Castro González.

2017 – 2024 **Bachelor's Degree in Energy Engineering**, Universidad Politécnica de Madrid

Experience

Nov 2025 – now **Predoctoral researcher**, TECF3IR group, Department of Energy Engineering, UNED, Madrid

- PhD on NACARTE, a CAD add-in that estimates photon dose rates with the point-kernel method, GPU ray tracing and neural networks (project PID2023-153160OA-I00). Migration from Ansys SpaceClaim to Ansys Discovery, ray tracing of the geometry around the dose map and comparison of neural network architectures.
- Training in ITER nuclear analysis within ITER Task Order #3, Roadmap Update 2025 (Nov 2025 – Jul 2026). Support to the integration of the Tokamak Cooling Water System into the ITER Full Model, and cleaning and simplification of CAD models converted to CSG with GEOUNED for the Mode-0 radiation maps.

Apr 2024 – Nov 2025 **AI Engineer**, Artificial Intelligence Department, HI Iberia, Madrid

- Differentiable deep learning surrogate models of the neutron transport in IFMIF-DONES, and multi-objective optimisation with them (DONES-FLUX project, MIP-20221017).
- Scalable software for HPC and data management on Kubernetes.

Jul 2023 – Apr 2024 **Research assistant**, Universidad Politécnica de Madrid

- Model of TVA Watts Bar Unit 1 in OpenMC, and a method to compute the neutron fluence in the extended beltline of the reactor vessel.

Apr 2022 – May 2023 **Simulation and modelling intern**, Tecnatom, Madrid

- Thermal-hydraulic system models and simulations with TEAM SUITE, and analysis and validation of ANSI tests for nuclear power plants.

Publications

1. **Differentiable Deep Learning Surrogate Models Applied to the Optimization of the IFMIF-DONES Facility**
   G. Gallardo Romero, G. Rodríguez-Llorente, L. Magariños Rodríguez, R. Morant Navascués, **N. Khvatkin Petrovsky**, R. Lorenzo Ortega, R. Gómez-Espinosa Martín
   Particles 8 (2025) 21. doi:10.3390/particles8010021

Conference contributions

1. **Accelerating and Optimizing Neutronic Simulations with Differentiable Neural Operator-based Surrogate Models**
   Talk, Fusion Neutronics Meeting 2025: ITER and beyond, UNED, Madrid, 7–10 April 2025.

2. **Deep Learning Surrogate Models for Neutronic Irradiation Optimisation in IFMIF-DONES**
   N. Khvatkin Petrovsky, L. Magariños Rodríguez, G. Rodríguez Llorente, R. Morant Navascués, G. Gallardo Romero, R. Lorenzo Ortega, R. Gómez-Espinosa Martín
   Poster, FISA 2025 & EURADWASTE '25, Warsaw, 12–16 May 2025.
3. **Aplicación de modelos subrogados con aprendizaje profundo para simulación neutrónica - proyecto DONES-FLUX**
   N. Khvatkin Petrovsky, L. Magariños Rodríguez, R. Morant Navascués, G. Rodríguez Llorente, R. Gómez-Espinosa Martín
   Talk, 50th Annual Meeting of the Spanish Nuclear Society (SNE), Córdoba, 9–11 October 2024.

Courses

Jun 2026	OpenMC workshop, 2nd Fusion Neutronics Meeting, KIT, Germany
Jun 2025	Introduction to Fusion Energy and Plasma Physics, Princeton Plasma Physics Laboratory
Jun 2025	Introduction to Kubernetes (LFS158), The Linux Foundation

Skills

Radiation transport	MCNP, D1SUNED, R2S, OpenMC, GEOUNED
Machine learning	PyTorch. Neural operators (DeepONet, MIONet, FNO), convolutional networks, Transformers
Programming	Python, C# (ILGPU, TorchSharp), Bash
Other tools	Ansys SpaceClaim and Discovery, HPC clusters with Slurm, Kubernetes, LaTeX, TEAM SUITE

Languages

Fluent	Spanish, Russian, English
Basic	German